RUNGTA INTERNATIONAL SKILLS UNIVERSITY

RUNGTA INTERNATIONAL SKILLS UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

PS No.	**84**
Project Title	**Hotel Revenue Analyzer**
Problem Statement	Read hotel_sales.csv with Pandas. Revenue = RoomsSold x Rate. Show daily revenue, the best day and the revenue share of each room type, with charts. Skip rows with negative or non-numeric values. **Language:** Python 3 **Concepts:** Pandas, CSV, Visualization **Note:** 2026-10-01,Standard,10,2000 **Note:** 2026-10-01,Deluxe,5,3500 **Note:** 2026-10-02,Standard,12,2000 **Note:** 2026-10-02,Deluxe,6,3500 **Note:** 2026-10-03,Standard,8,2000 **Note:** 2026-10-03,Deluxe,4,3500 **Sample data (4 columns, 0 rows):** `Date,RoomType,RoomsSold,Rate`
Test Cases / Case Statements (4 cases)	See table below
Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Load the CSV file with Pandas and report how many valid rows were read. 3. Create a Revenue column (RoomsSold x Rate). 4. Skip rows with negative or non-numeric values and show a warning. 5. Test the program with all four test cases.
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
File: hotel_sales.csv (sample above) Menu: Daily revenue	2026-10-01 : Rs. 37500 2026-10-02 : Rs. 45000 2026-10-03 : Rs. 30000 Total revenue: Rs. 112500 Best day: 2026-10-02 (Rs. 45000)
Test Case 2: Boundary / Minimum Input	
File with one data row: 2026-10-01,Standard,1,2000 Menu: Daily revenue	2026-10-01 : Rs. 2000 Total revenue: Rs. 2000 Best day: 2026-10-01 (Rs. 2000)
Test Case 3: Invalid Input	
(a) File: hotel_sales.csv with an extra row 2026-10-04,Deluxe,-3,3500 (b) File: sales_old.csv (does not exist)	(a) Warning: Row 7 skipped - RoomsSold must be 0 or more. Valid records analysed: 6 (b) Error: File 'sales_old.csv' not found.
Test Case 4: Different Valid Scenario	
File: hotel_sales.csv (sample above) Menu: Revenue by room type	Standard : Rs. 60000 (53.33%) Deluxe : Rs. 52500 (46.67%) Total revenue: Rs. 112500

```
PS C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer> & 'c:\python315\python.exe' 'c:\Users\prosi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '59150' '--' 'C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer\hotel_revenue_analyzer.py'
========================================
      HOTEL REVENUE ANALYZER
========================================
Enter CSV filename [hotel_sales.csv]: hotel_sales.csv
Valid records analysed: 6

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 1

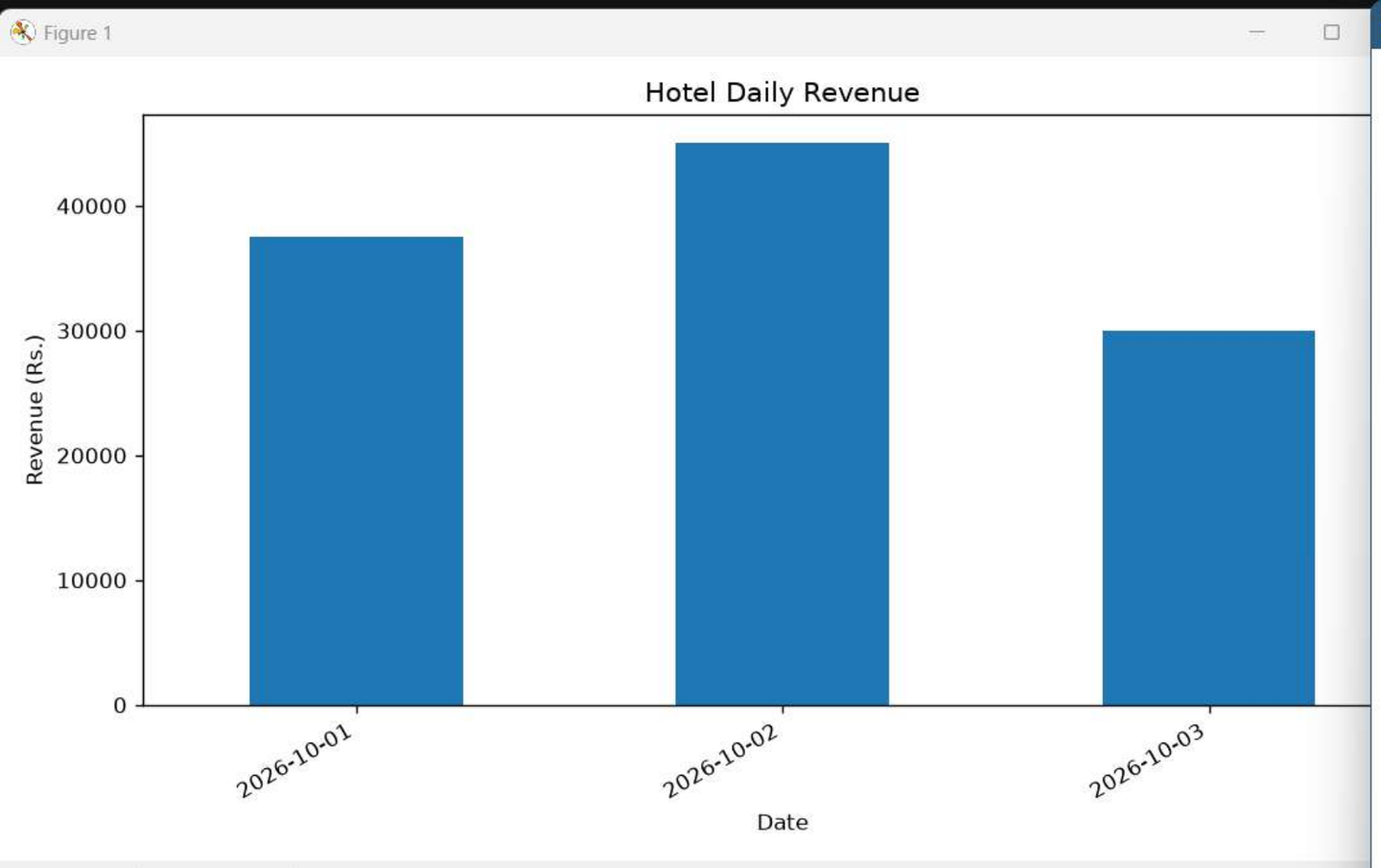
--- DAILY REVENUE ---
2026-10-01 : Rs. 37500
2026-10-02 : Rs. 45000
2026-10-03 : Rs. 30000

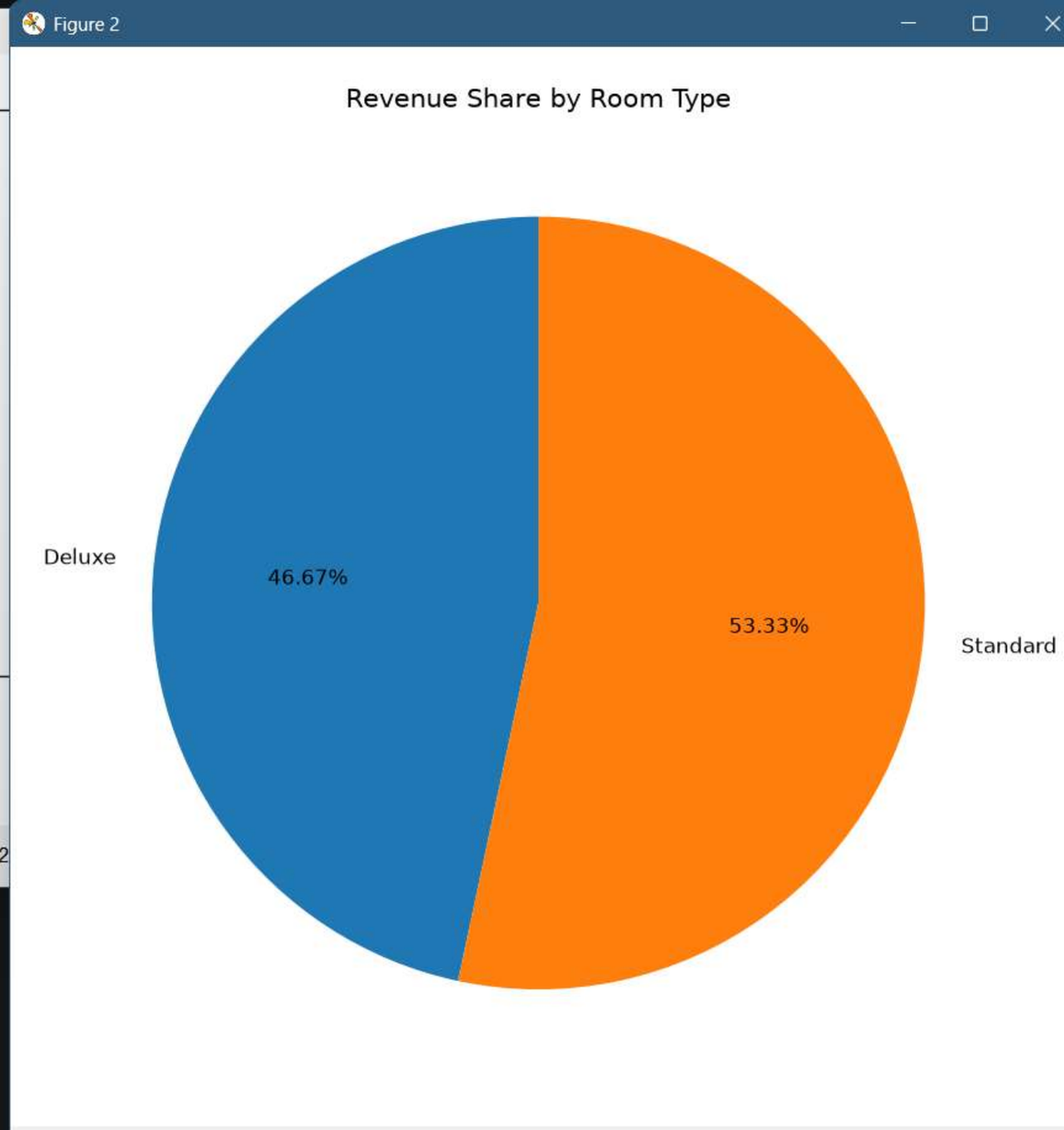
Total revenue: Rs. 112500
Best day: 2026-10-02 (Rs. 45000)

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 2

--- REVENUE BY ROOM TYPE ---
Deluxe : Rs. 52500 (46.67%)
Standard : Rs. 60000 (53.33%)
Total revenue: Rs. 112500

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 3
```





(x, y) = (, 2.72

```
Enter your choice (1-4): 2

--- REVENUE BY ROOM TYPE ---
Deluxe : Rs. 52500 (46.67%)
Standard : Rs. 60000 (53.33%)
Total revenue: Rs. 112500

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 3
```

```
PS C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer> & 'c:\python315\python.exe' 'c:\Users\prosi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '50817' '--' 'C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer\hotel_revenue_analyzer.py'
Enter CSV filename [hotel_sales.csv]: hotel_sales.csv
Valid records analysed: 6

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 1

--- DAILY REVENUE ---
2026-10-01 : Rs. 37500
2026-10-02 : Rs. 45000
2026-10-03 : Rs. 30000

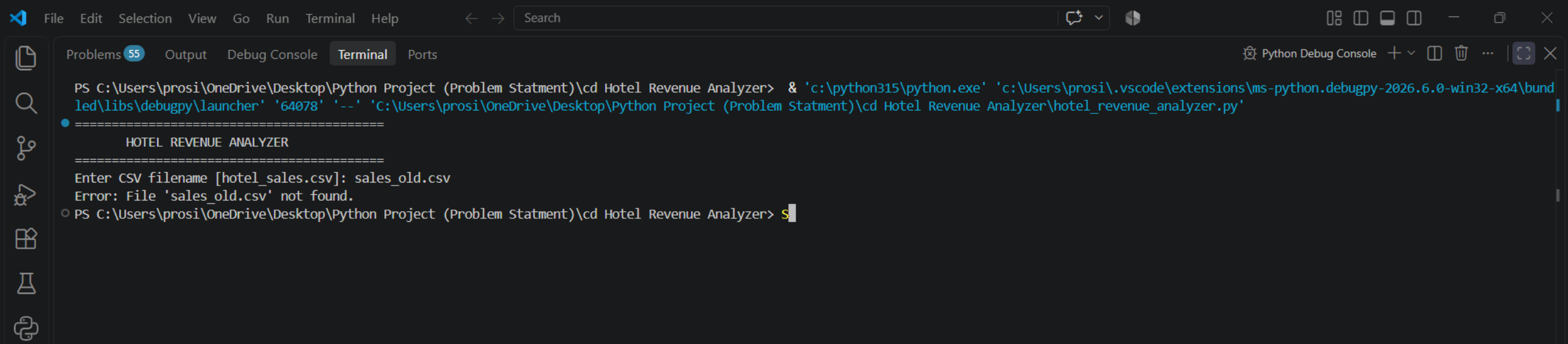
Total revenue: Rs. 112500
Best day: 2026-10-02 (Rs. 45000)

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 2

--- REVENUE BY ROOM TYPE ---
Deluxe : Rs. 52500 (46.67%)
Standard : Rs. 60000 (53.33%)
Total revenue: Rs. 112500

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 3

========== MAIN MENU ==========
1. Daily Revenue and Best Day
2. Revenue by Room Type
3. Display Charts
4. Exit
Enter your choice (1-4): 4
Thank you for using Hotel Revenue Analyzer!
PS C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer>
```


File Edit Selection View Go Run Terminal Help
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Problems 55 Output Debug Console Terminal Ports
Python Debug Console
PS C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer> & 'c:\python315\python.exe' 'c:\Users\prosi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '64078' '--' 'C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer\hotel_revenue_analyzer.py'
==========
HOTEL REVENUE ANALYZER
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Enter CSV filename [hotel_sales.csv]: sales_old.csv
Error: File 'sales_old.csv' not found.
PS C:\Users\prosi\OneDrive\Desktop\Python Project (Problem Statment)\cd Hotel Revenue Analyzer> S